

HeatWatch AI

AI-Based Climate Intelligence System for Heatwave Monitoring, Prediction, and Early Warning

Historical Data Report

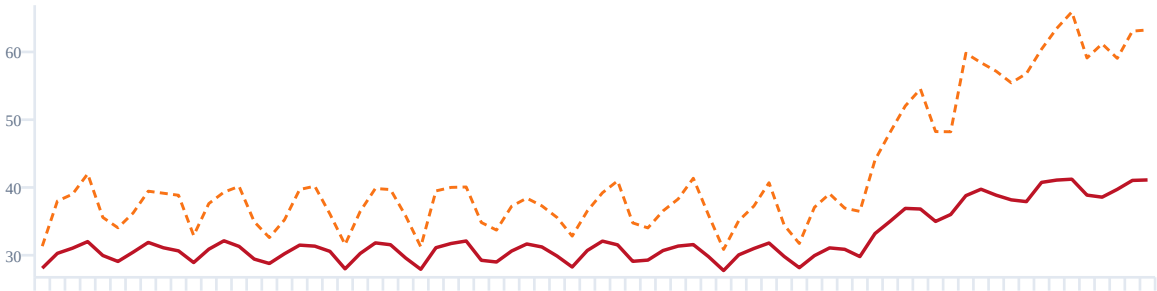
LOCATION	REPORTING PERIOD	GENERATED
Mumbai, Maharashtra	2026-09-05 to 2026-09-19	19 Sep 2026, 13:03 UTC · display tz Asia/Kolkata
COORDINATES	DATA PROVENANCE	UNITS
19.0760, 72.8777	DEMO	°C · km/h · hPa

1 · Weather Statistics

Metric	Value	Metric	Value
Observations stored	74	Avg temperature	32.41 °C
Max temperature	41.22 °C	Min temperature	27.74 °C
Avg humidity	69.74 %	Avg wind speed	13.62 km/h
Heat index (avg)	41.92 °C	Heat index (max)	65.9 °C

Heat index is reported only where the NWS Rothfusz calculation is applicable (74 of 74 observations); otherwise air temperature is used as the thermal-stress proxy and noted as unavailable.

Temperature trend (crimson) and heat index (dashed orange) — stored observations



2 · Heatwave Risk Analysis

Predictions	15
Low risk	0
Moderate risk	11
High risk	4
Average severity	69.81 / 100
Peak severity	100.0 / 100

Risk assessments combine the weather-based severity model with a documented, bounded sentiment-fusion layer (upward-only, max +12). Methodology and model version are recorded with every prediction; see the HeatWatch AI model card (docs/model/model_card.md) for evaluation scope and limitations.

3 · Sentiment Analysis Summary

Posts analysed	142
Positive	71
Neutral	32

Negative	39
Heat-related posts	61
Heat-related distress signals	30
Average sentiment score	0.12

Sentiment model: VADER (pretrained, English social-media lexicon; compound score in [-1, 1]). General sentiment is distinct from heat-related distress: the distress count includes only heat-relevant posts with health/impact vocabulary. Sentiment is a complementary signal, not a standalone severity measure.

4 · Alert Summary

GENERATED (UTC)	LEVEL	SEVERITY	STATUS
2026-09-16 10:00	HIGH	100/100	acknowledged
2026-09-17 10:00	HIGH	100/100	acknowledged
2026-09-18 10:00	HIGH	100/100	active
2026-09-19 10:00	HIGH	98/100	active

Total alerts in period: 4 (high: 4, moderate: 0). All alerts are application-generated early warnings with cooldown and escalation policy — they are not official government advisories.

5 · Data Source and Methodology Notes

- **Data provenance: DEMO** — records marked DEMO are seeded demonstration data and do not represent real observations or real public posts.
- Weather observations: provider(s) DEMO; units: temperature in degC, wind in km/h, pressure in hPa.
- Heat index: NWS Rothfusz regression with adjustments; reported only where applicability conditions hold.
- Sentiment model: VADER (pretrained, English social-media lexicon), compound score in [-1, 1].
- Severity model: weather-based scikit-learn RandomForest trained on the documented heat-index banding dataset, combined with a rule-based, bounded sentiment fusion layer. See the model card for evaluation metrics and limitations.
- Report generated by HeatWatch AI; every figure is derived exclusively from the database records matching the selected filters.